

India's Natural Resources and Its Industrialization

Zhihu column · 2026-07-03

Recently I have seen many people argue that India cannot industrialize, citing its resource poverty: little oil, little gas, poor coal, few minerals, not even enough water. Without energy, without electricity, without industrial raw materials — how can industry ever develop?

In this article, taking China's level around 2012 as the benchmark for having essentially completed industrialization, we actually analyze whether India's natural resources can support its own industrialization, and whether a lack of natural resources will prevent it from completing the task.

I. Energy Resources

Below is a comparison of the key primary energy resources. Among fossil fuels, India indeed has very little oil and gas, but its coal resources are abundant. Among non-fossil energy, India's thorium resources are extremely rich, and the conditions for developing nuclear power are also very good.

Resource category	China	India	Ratio
Oil (proven reserves)	26 billion barrels	5.7 billion barrels	4.6:1
Natural gas (proven reserves)	8.4 trillion cu m	1.4 trillion cu m	8.4:1
Coal (proven reserves)	249.2 billion tonnes	111.052 billion tonnes	2.2:1
Uranium (proven reserves)	350,000 tonnes	170,000 tonnes	2:1
Thorium (proven reserves)	280,000 tonnes	840,000 tonnes	1:3
Hydropower (technically exploitable)	687 GW	149 GW	4.6:1
Solar (mean annual GHI)	1400–1500 kWh/m ²	1800–2300 kWh/m ²	1:1.4
Wind (technically exploitable)	2,700 GW	1,164 GW	2.3:1

1. Oil and Gas

India's poverty in oil and gas does pose a certain difficulty for its industrial development, but India's geographic position is very good: it lies close to the Persian Gulf and can rely on imports. Let us analyze the cost of imported energy when India essentially completes its industrialization by 2040 (at roughly China's 2012 level).

- India's current annual crude oil output is about 30 million tonnes, and its annual consumption about 240 million tonnes (roughly China's level in 2005), relying essentially on imports. By 2040, India's annual crude consumption is projected to surge to about 500 million tonnes (about 10 million barrels per day), comparable to China's 2017 level of consumption. Although the Indian government has plans to extract deep-sea oil ^[1] ^[2], the journey from exploration to production takes a long time; probably not until after 2040. So, if India's domestic output remains at around 30 million tonnes, it will need to import at least 470 million tonnes of crude a year, at a total import cost of roughly \$200 billion a year.
- India's current annual natural gas output is 35 billion cubic meters, against consumption of 65 billion cubic meters (about China's 2007 level), requiring imports of 30 billion cubic meters. By 2040, India's annual gas consumption is projected to rise to 113.7 billion cubic meters ^[3], a little above China's 2010 level. In 2010, China's total gas output was 94.48 billion cubic meters, and its total annual gas imports (LNG plus pipeline gas) were about 16 billion cubic meters. No projection of India's gas output in 2040 was found; domestic output is not expected to exceed 50 billion cubic meters, so at least 60 billion cubic meters will have to be imported, at a total cost of around \$20 billion a year.

Thus in 2040 India will need to import oil and gas worth about \$220 billion a year (in 2026 dollars) — roughly double the current oil-and-gas import bill of some \$120 billion. By comparison, China spent \$135.1 billion on crude oil imports in 2010 ^[4], equivalent to about \$200 billion in 2026 dollars. In fact, India's import volume is not that much larger — this is mainly the benefit of electrification. Taking a middle estimate, from 2025 to 2040 India's average annual oil-and-gas imports will be roughly \$170 billion.

2. Electricity

India's solar resource is superb. In the Thar Desert of Rajasthan, for example, the mean annual global horizontal irradiation (GHI) is about 2000 to 2200 kWh/m²; China's Taklamakan Desert, by comparison, receives 1500 to 1800 kWh/m² — a notch lower. India is now vigorously developing solar power in Rajasthan, and there have already been cases of "solar plants devouring people" ^[5].

Under the Indian government's plans, by 2030 it will complete 500 GW of non-fossil capacity, most of it photovoltaic ^[6]. By 2040, India's PV capacity is expected to reach 700 GW. Given Rajasthan's insolation, 700 GW of PV could generate 1,100 billion kWh a year — more than 900 kWh per capita for India.

India's PV industry is growing very fast. According to media reports, "in 2025 India's PV module manufacturing capacity surged by 119 GW, cell capacity grew by over 9 GW, and domestic

manufacturing capacity expanded rapidly" [7]. The government's local-content policy is being tightened further: "from June 1, some Indian PV projects must use wholly domestically manufactured solar cells" [8], and for wafers, "from June 1, 2028, all solar projects, including net-metered and open-access ones, must use wafers on the ALMM list" [9]. The Indian authorities' stance is extremely firm — clearly determined by the real situation of the domestic industry and the strategic position of photovoltaics. PV has become a key pillar of India's industrialization. It is foreseeable that what India imports from China will gradually shrink to little more than production equipment. As the share of local production keeps rising, and taking an average of \$100 million in equipment and materials imports per GW, the 700 GW of PV will require \$700 billion in total. Spread over fifteen years, that is about \$4.6 billion a year — more in the early years, less later.

PV needs a certain share of energy storage, to shift surplus daytime electricity to the evening peak (6 to 10 p.m.). At a 25 percent ratio, $700 \text{ GW} \times 0.25 = 175 \text{ GW}$ of 4-hour storage is needed; if all of it is chemical (battery) storage, the total cost is roughly \$50 billion. Pumped-storage hydropower is costlier to build but lasts longer, and on the whole is cheaper. And the civil engineering of pumped storage is certainly within the capacity of Indian construction firms — at most the equipment needs importing. So we can simply estimate a uniform \$50 billion. India is now vigorously promoting localization of battery production [10], and Chinese firms have entered the Indian market in force, cooperating with local companies [11]. Actual foreign-exchange outlay will therefore be well below \$50 billion. Spread over fifteen years, about \$3.3 billion a year, again more early and less later.

Since PV produces mainly by day, the nighttime base load still has to be carried by coal and nuclear power. India currently has about 230 GW of coal-fired capacity; another 100 to 200 GW is needed to satisfy domestic demand. Taking 200 GW, the total cost is roughly \$200 billion. As India already has the domestic capability for turnkey thermal plants and most components, the import share is low — say 20 percent, or \$40 billion. Spread over fifteen years, that is a little over \$200 million a year. Indian coal is of poorer quality and a large part must be imported, but the share of domestic coal has risen sharply recently, reaching 50 percent [12]. India currently imports over \$20 billion worth of coal a year; as coal plants multiply, imports are expected to rise to around \$35 billion a year.

India's nuclear capacity is now close to 10 GW, with a plan to increase it to 100 GW by 2047. On that ratio, by 2040 nuclear capacity will be about 70 GW. The main reactor type India plans to deploy — the PHWR (pressurized heavy water reactor) — has essentially been localized, reducing imports; the exact proportion is unknown and we will not consider it here. But India's uranium output is low, and uranium must be imported: 100 GW of nuclear power needs about 20,000 tonnes of uranium a year, costing close to \$4 billion. After the 2050s, however, India's thorium reactors should be technically ready — India has the world's largest thorium reserves, and thorium reactors will be promoted vigorously thereafter. By 2040, India's coal and nuclear capacity together will reach about 500 GW,

solar 700 GW, plus about 200 GW of storage; without counting wind, hydro, and the rest, this essentially reaches China's 2015 level. The annual import costs for construction and fuel of the main generating types above amount to about $4.6 + 3.3 + 35 + 4 = \$46.9$ billion.

3. Summary of Energy

Annual oil-and-gas costs plus other generating-plant construction costs come to \$170 billion + \$46.9 billion = \$216.9 billion. In 2025, India imported about \$120 billion of oil and gas, about \$4 billion of PV equipment ^[13], and about 250 million tonnes of coal ^[14] costing roughly \$27 billion — about \$151 billion in all. In other words, over the next fifteen years India will need to import an additional \$65.9 billion a year to support its industrialization.

II. Metals and Minerals

The bulk commodities of the global market are mainly iron, copper, aluminum, zinc, and gold, plus the new-energy-related lithium and nickel. These account for over 90 percent of global metals-and-minerals trade, with iron and copper the largest. Below we examine India's situation in detail.

1. Iron Ore

India has large, rich, potential iron ore reserves of over 20 billion tonnes. Hematite accounts for 75 percent of total reserves, and more than half of the hematite is medium- to high-grade ore with over 62 percent Fe content — the kind that can go directly into blast furnaces as lump, sinter, or pellet ^[15]. That means India alone has 7.5 billion tonnes of ore at 62 percent iron content, enough to smelt 4.8 billion tonnes of crude steel. Of course, India can also import ore from Brazil and Australia. India's strategy is to export low-grade ore and import high-grade ore, with exports overall exceeding imports ^[16]. By contrast, China's iron ore is of very low grade ^[17] — far inferior to India's. Hence China must import vast quantities of ore every year for its steel production. For example, "in 2010 China imported 619 million tonnes of iron ore in all, down about 9 million tonnes from 2009; in value terms, 2010 imports cost about \$78.9 billion." On the whole, India's iron ore resources are ample enough for its own industrialization.

2. Copper

India has very little copper ore — negligible — and its copper needs must basically be imported. China consumed about 54 million tonnes of copper from 2004 to 2012. If India needs roughly the same amount of copper to complete its industrialization, it will have to import about 50 million tonnes of

copper or copper ore. For convenience, counting it all as copper, the total price is about \$650 billion — i.e., \$43.3 billion a year over fifteen years.

3. Bauxite

China consumed about 100 million tonnes of aluminum from 2004 to 2012. If India's industrialization requires the same amount, it will need 500 million tonnes of bauxite. India's proven bauxite reserves are 650 million tonnes ^[18], fully sufficient for industrialization. And since India's bauxite resources total more than 3 billion tonnes ^[19], further exploration will turn up new deposits — bauxite should not become a bottleneck for India.

4. Gold

In 2025, China's industrial use of gold was 82 tonnes ^[20], worth about \$10 billion. India's domestic gold production is negligible, but its private-sector holdings of gold exceed 20,000 tonnes ^[21]; purely in resource terms, its domestic gold is quite enough for its own industrialization. Here we simply count \$10 billion a year in gold imports for India.

5. Zinc

India's zinc resources amount to more than 30 million tonnes ^[22], of which about 10 million tonnes are proven reserves ^[23]. China consumed about 43 million tonnes of zinc from 2004 to 2012. If no new zinc resources are found in India, it must import 13 million tonnes, at a total cost of roughly \$40 billion — about \$2.7 billion a year over fifteen years.

6. Lithium

India's lithium reserves are small; we set them aside. China's lithium consumption in 2012 was very low; with the development of the new-energy industry, consumption in the last couple of years has reached about 1 million tonnes. Suppose India's average consumption over the fifteen years from 2026 to 2040 reaches half of China's — 500,000 tonnes — then the annual cost averages about \$10 billion.

7. Nickel

India's nickel reserves are also small; set aside. India's current nickel consumption is only a few tens of thousands of tonnes; suppose it averages 500,000 tonnes a year over the next fifteen years, then the annual cost averages about \$8 billion.

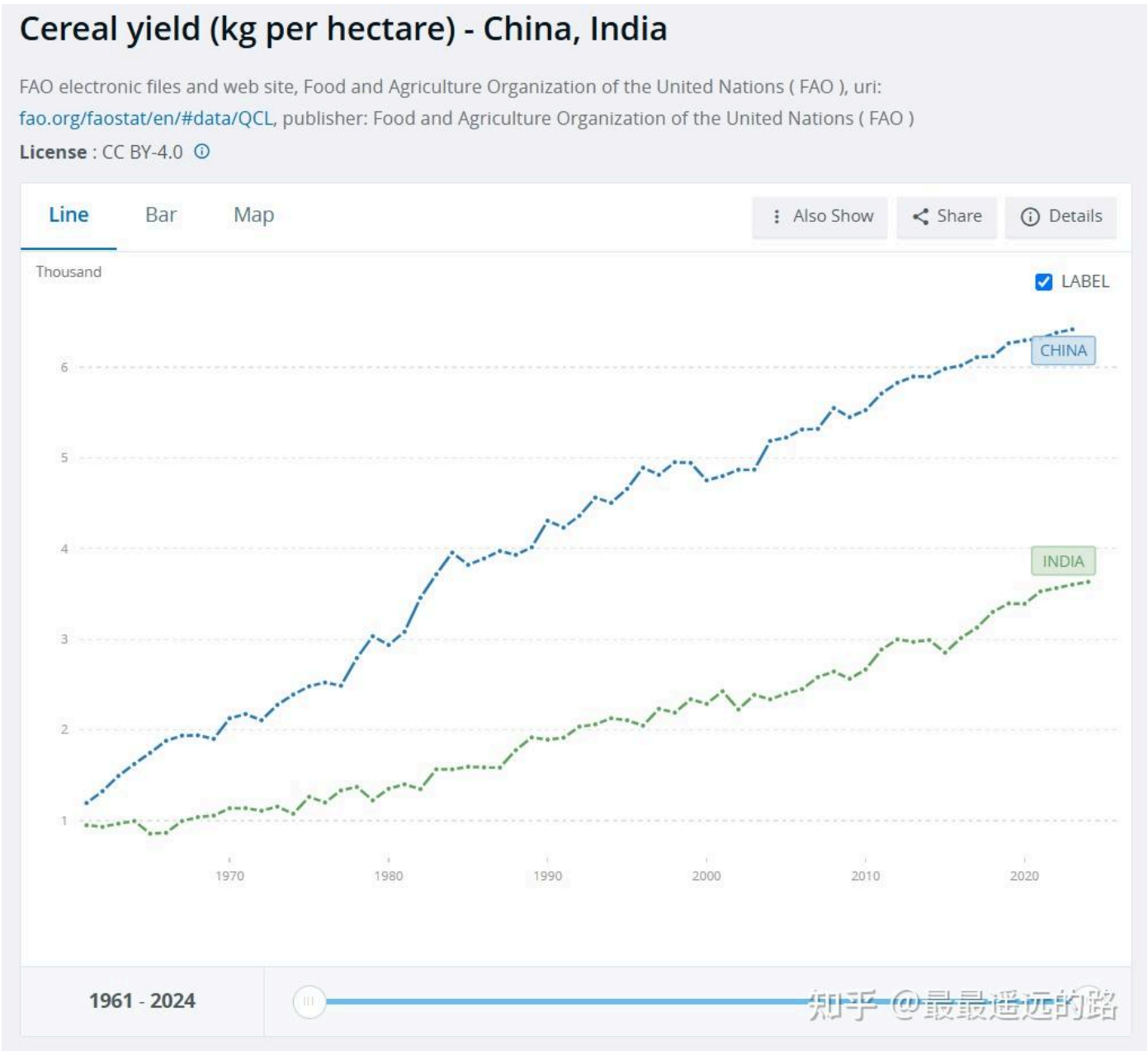
8. Summary of Metals and Minerals

Annual imports of metals and minerals total \$76 billion (\$43.3 billion copper + \$2.7 billion zinc + \$10 billion gold + \$10 billion lithium and \$8 billion nickel). India's current gold imports alone amount to \$71.9 billion — that single item can offset all of its metals-and-minerals needs.

III. Other Resources

1. Farmland

Although India's land area is smaller, its arable land exceeds China's [24]. India's grain output is low not for lack of land but for lack of inputs such as fertilizer. As India's basic industrial strength grows, its per-hectare grain yields will gradually rise. India's grain yields have already reached China's level of the 1980s [25].



Likewise, India's fertilizer application per unit area has also reached China's 1980s level ^[26]. There is a high correlation between grain output and fertilizer use. And with green ammonia technology, cheap solar and wind power can now be used to make ammonia fertilizer — so technically, there is no resource constraint for India here.

Fertilizer consumption (kilograms per hectare of arable land) - India, China

FAO electronic files and web site, Food and Agriculture Organization of the United Nations (FAO), publisher: Food and Agriculture Organization of the United Nations (FAO)

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2. Water Resources

India's total precipitation is 3,880 BCM (billion cubic meters); owing to topography, evaporation, and other constraints, its actually utilizable water (renewable freshwater resources) is about 1,122 BCM ^[27]. Of this, 690 BCM is surface water and 436 BCM is annual groundwater recharge.

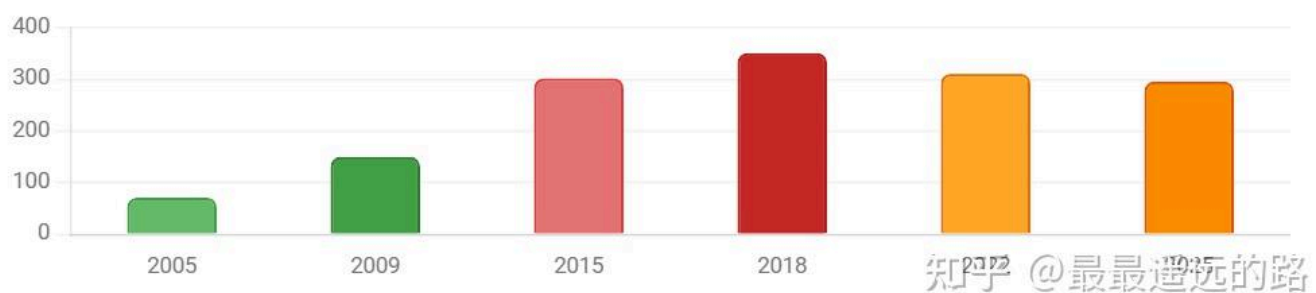
India's current annual water use is 761 BCM, still less than 1,122 BCM. China, by comparison, uses about 600 BCM a year, with far higher efficiency. India's water difficulty therefore lies not in shortage of water but in its mode and efficiency of use. The main causes are several:

1) Rice is planted over a wide area, and irrigation is inefficient. India grows rice not only in high-rainfall regions but in states with little available water, so groundwater tables fall year after year [28]. For example, Punjab's average annual available water is about 35 BCM — around 14.8 BCM of surface water and 18.6 BCM of annual groundwater recharge, of which 16.8 BCM is extractable [29]. But the province's total annual social water use runs to 28–34 BCM, over 90 percent for agricultural irrigation, most of it for rice. In 2025, the province extracted 26.27 BCM of groundwater — far exceeding recharge. Several southern states are in a similar situation: rainfall is a little higher, but storage capacity is insufficient to make effective use of the heavy monsoon rains.

2) Surface water is badly polluted, rendering large quantities unusable. Because much industrial and domestic sewage is discharged directly into rivers, the river water is too polluted to serve as a source for industry or agriculture. As the figure below shows [30], India's river pollution worsened with industrialization and urbanization, peaking in 2018. It has improved somewhat in recent years, but remains very serious. India's surface water (mainly rivers and lakes) totals 690 BCM, of which 70 percent is polluted and only 30 percent usable. By contrast, over 90 percent of China's surface water is usable [31].

Polluted River Stretches in India (2005–2025)

Source: [CPCB National Water Quality Monitoring Programme](#), various assessment years



In 2005, the Central Pollution Control Board's monitoring identified 71 polluted river stretches. A decade later the number had more than quadrupled. By 2018, assessments recorded 351 polluted stretches across 323 rivers — an all-time high. The 2022 assessment showed a slight decline, with 311 stretches across 279 rivers; the latest 2025 report further reduced the figures to 296 stretches across 271 rivers

3) The rate of sewage treatment and reuse is low, wasting water resources. China's use of reclaimed water in 2025 was 22 BCM ^[32]; India's total urban sewage is 26 BCM, of which only 28 percent is treated, and the reuse rate of reclaimed water is under 3 percent — essentially negligible. Since industrial water use is concentrated in cities, if reclaimed water is not widely used by industrial enterprises, urban water scarcity easily results.

Serious as these three problems are, they are not insoluble.

First, India can cut agricultural water use by raising irrigation efficiency. Indian irrigation is inefficient: on the same area with the same crop, China uses only about half as much water. India's roughly 700 BCM of annual agricultural water use could be reduced to about 400 BCM, freeing up 300 BCM of water. In addition, local policies encouraging farmers to switch crops have already begun to take effect ^[33]: by subsidizing land planted to less thirsty crops, farmers are steered toward lower water use. This policy resembles China's earlier Grain-for-Green program.

Second, sewage treatment and new storage infrastructure can increase usable surface water. China is a pioneer here. In 2005, about 60 percent of China's surface water was too polluted to use ^[34]; by 2015 that share had fallen to about 35 percent ^[35]. If the polluted share of India's surface water falls to 40 percent, more than 200 BCM of surface water becomes usable. Moreover, with extensive restoration and construction of ponds, reservoirs, wetlands, and other storage — the official Indian estimate — another 185 BCM of water can be added ^[36]. Indian officialdom and civil society are both acting on this.

Third, urban water stress can be eased by strengthening sewage treatment and the recycling of reclaimed water. If all sewage were treated and reused, another 26 BCM would be available.

All told, India can free up and add $300 + 200 + 185 + 26 = 711$ BCM of water. That already exceeds China's entire annual water use — for the people of India, it must be enough.

IV. Conclusion

On our analysis above, if India is to reach roughly China's 2012 level of industrialization by 2040, then in addition to its domestic resources it must spend, over the next fifteen years, an extra \$65.9 billion a year on oil-and-gas imports and \$76 billion a year on metals-and-minerals imports — about \$141.9 billion a year in all. Where does this foreign exchange come from? On one side, exports increase — a large part of the rise in imports is for the sake of exports. On the other side, remittance income rises. India's remittances in 2025 were \$135.5 billion, versus only \$68.9 billion in 2015 — nearly doubling in a decade. Indians are now streaming abroad to work in vast numbers, and this growth rate can be sustained roughly through 2040. By 2035, remittances are projected at \$230–250

billion, an increase of at least \$100 billion. So the foreign exchange India needs for industrialization should not be a major problem.

Moreover, India's water problem does not constitute a physical bottleneck either: by treating pollution, adding storage facilities, and reducing agricultural water use, the shortage can be effectively eased, even essentially resolved.

Therefore, India faces no problem of insufficient resources for industrialization. By around 2040 it can reach roughly China's 2012 level and essentially complete its industrialization.

Notes

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[33] Haryana Government Announces ₹8,000/Acre Incentive for Farmers Shifting from Paddy to Alternative Crops

中文原文：Haryana Government Announces ₹8,000/Acre Incentive for Farmers Shifting from Paddy to Alternative Crops

https://www.global-agriculture.com/india-region/haryana-government-announces-%E2%82%B98000-acre-incentive-for-farmers-shifting-from-paddy-to-alternative-crops/#google_vignette

[34] 《2005环境状况公报》国家环境监测网(简称国控网)七大水系的411个地表水监测断面中，I~III类、IV~V类和劣V类水质的断面比例分别为41%、32%和27%。其中，珠江、长江水质较好，辽河、淮河、黄河、松花江水质较差，海河污染严重。主要污染指标为氨氮、五日生化需氧量、高锰酸盐指数和石油类。

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<https://news.sina.cn/sa/2006-06-15/detail-ikkntiam7217476.d.html>

[35] 《2015环境状况公报》全国967个地表水国控断面（点位）开展了水质监测，I~III类、IV~V类和劣V类水质断面分别占64.5%、26.7%和8.8%。

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<https://www.mee.gov.cn/hjzl/sthjzk/zghjzkgb/201606/P020160602333160471955.pdf>

[36] 《MASTERPLAN FOR ARTIFICIAL RECHARGE OF GROUNDWATER》Master Plan for Artificial Recharge to Groundwater- 2020 has been prepared by Central Ground Water Board (CGWB) in consultation with States/UTs which is a macro level plan indicating various GW recharge structures for the different terrain conditions of the country. The plan aims to construct approximately 1.42 crore such structures across India, potentially harnessing about 185 Billion Cubic Meters (BCM) of rainwater.

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<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147822®=48&lang=2>

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